Inorganic Cumulative Exam

2/15/13

Given the diversity of compounds that fall in the category of inorganic chemistry it is not surprising that a number of methods have been developed to help determine the structure or the stoichiometry of a compound or class of compounds. Below are some questions that make use of these methods.

1. Describe and draw the structure of B_5H_9

2. Describe and draw the structure of B_4H_{11}

3. Describe and draw the structure of ICl_3

4. Describe and draw the structure of SO_3

5. Describe and draw the structure of XeF_4

6. Describe and draw the structure of 1,2-$B_{10}C_2H_{12}$

7. Describe and draw the structure of cubic ZnS

8. Give the formula for the simplest manganese carbonyl compound (i.e., what are x and y in Mn_xCO_y)

9. Give the formula for the simplest compound made from benzene and Cr(O)

10. Describe and draw the structure of $P_3O_{10}^{5-}$.